M378K Introduction to Mathematical Statistics
Problem Set #14
Statistics.

Definition 14.1. *A* random sample *of size n from distribution D is a random vector*

$$(Y_1, Y_2, \ldots, Y_n)$$

such that

1. *$Y_1, Y_2, \ldots, Y_n$ are independent, and*
2. *each Y_i has the distribution D.*

Example 14.2. **Quality control**. *Times until a breaker trips under a particular load are modeled as exponential. The intended procedure is to choose n breakers at random from the assembly line, subject them to the load, and measure the time it takes for them to trip. The lifetime of a specific breaker indexed by i is a random variable Y_i with an exponential distribution with an unknown parameter $\theta = \tau$. Independence of Y_i, $i = 1, \ldots, n$ is assured by the random choice of breakers to test.*

Definition 14.3. *A* statistic *is a function of the (observable) random sample and known constants.*

Problem 14.1. *Give at least three examples of statistics of a certain random sample $Y_1, Y_2, \ldots, Y_n$.*

- $\bar{Y} = \frac{1}{n}(Y_1 + \cdots + Y_n)$ ... sample mean
- $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$ ... sample variance
- $Y_{(n)} = \max(Y_1, \ldots, Y_n)$

Remark 14.4. *Statistics are random variables in their own right. We call their probability distributions* sampling distributions.

Example 14.5. **Quality control, cont'd.** *Let the random variable Y be the minimum of random variables $Y_1, \ldots, Y_n$, i.e., the shortest time until the breaker is tripped in the sample. We can write*

$$Y = \min(Y_1, \ldots, Y_n).$$

What is another name for this random variable? First order statistic.

Then, the sampling distribution *of Y can be figured out by looking at its cumulative distribution function. We have …*

$y>0$: $F_{(1)}(y) = \mathbb{P}[\min(Y_1,\dots,Y_n) \le y] = 1-\mathbb{P}[\min(Y_1,\dots,Y_n) > y] =$

$= 1-\mathbb{P}[Y_1 > y, Y_2 > y, \dots, Y_n > y] =$ independence

$= 1-\mathbb{P}[Y_1 > y]\cdot\mathbb{P}[Y_2 > y]\cdots\mathbb{P}[Y_n > y]$

$= 1-(1-F_Y(y))^n = 1-(\cancel{1}-(\cancel{1}-e^{-\frac{y}{\tau}}))^n = 1-e^{-\frac{y}{\tau/n}} \Rightarrow \boxed{Y_{(1)} \sim E\left(\frac{\tau}{n}\right)}$

Problem 14.2. *Let $Y_1, \dots, Y_n$ be a random sample of size n from a normal distribution with mean μ and variance σ^2. What is the sampling distribution of*

$$\bar{Y}_n = \frac{1}{n}\sum_{k=1}^{n} Y_k \quad ?$$

$$\boxed{\bar{Y}_n \sim \text{Normal}\left(\text{mean} = \mu,\ \text{sd} = \frac{\sigma}{\sqrt{n}}\right)}$$

Estimators.

Def'n. The bias of an estimator $\hat{\theta}$ of the parameter θ is defined as:

$$\text{bias}(\hat{\theta}) := \mathbb{E}[(\hat{\theta}-\theta)] = \mathbb{E}[\hat{\theta}]-\theta$$

Notation from books: "$\mathbb{E}_\theta(\cdot)$, $\mathbb{E}^\theta(\cdot)$, $\mathbb{E}[\cdots|\theta]$"

We say that an estimator $\hat{\theta}$ is unbiased for the parameter θ if

$$\boxed{\text{bias}(\hat{\theta})=0 \iff \mathbb{E}[\hat{\theta}]=\theta}$$

for all admissible values of θ.

Example. Consider a random sample $Y_1,\dots,Y_n$ from $N(\mu,\sigma)$ w/ both μ and $\sigma>0$ unknown

$$\hat{\mu} = \bar{Y} := \frac{1}{n}(Y_1+\dots+Y_n)$$

Then, $\mathbb{E}[\hat{\mu}]=\mu$, i.e., $\hat{\mu}=\bar{Y}$ is unbiased for μ.

Example. Let $Y_1,\dots,Y_n$ be a random sample from $N(\mu_0,\sigma)$ w/ μ_0 known and $\sigma>0$ unknown.

We propose this estimator for the variance σ^2:

$$\boxed{S^2 := \frac{1}{n}\sum_{i=1}^{n}(Y_i-\mu_0)^2}$$

Then,

$$\mathbb{E}[S^2] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(Y_i-\mu_0)^2\right]$$

$$= \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(Y_i-\mu_0)^2\right]$$

$$= \frac{1}{n}\sum_{i=1}^{n}\underbrace{\boxed{\mathbb{E}[(Y_i-\mu_0)^2]}}_{\sigma^2} = \frac{1}{\cancel{n}}\cdot\cancel{n}\cdot\sigma^2 = \sigma^2$$

$\Rightarrow$ S^2 is unbiased for σ^2

Example. Let $Y_1, \dots, Y_n$ be a random sample from $N(\mu, \sigma)$ w/ both μ and σ unknown.

Goal: Find a "good" estimator for σ^2!

Propose:

$$(S')^2 := \frac{1}{n} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

Q: Is $(S')^2$ unbiased for σ^2?

$$\mathbb{E}[(S')^2] = \frac{1}{n} \sum_{i=1}^{n} \boxed{\mathbb{E}[(Y_i - \bar{Y})^2]}$$

$$\mathbb{E}[Y_i^2 - 2 \cdot Y_i \cdot \bar{Y} + \bar{Y}^2]$$

$$= \frac{1}{n} \left(\sum_{i=1}^{n} \mathbb{E}[Y_i^2] - 2 \sum_{i=1}^{n} \mathbb{E}[Y_i \cdot \bar{Y}] + \overbrace{\boxed{\sum_{i=1}^{n} \mathbb{E}[(\bar{Y})^2]}}^{n \cdot \mathbb{E}[(\bar{Y})^2]} \right)$$

$$= \frac{1}{n} \left(n \cdot \mathbb{E}[Y_1^2] - 2 \mathbb{E}[\overbrace{\sum_{i=1}^{n} Y_i}^{n \cdot \bar{Y}} \bar{Y}] + n \cdot \mathbb{E}[(\bar{Y})^2] \right)$$

$$= \frac{1}{\cancel{n}} \left(\cancel{n} \cdot \mathbb{E}[Y_1^2] - 2 \cancel{n} \, \mathbb{E}[(\bar{Y})^2] + \cancel{n} \cdot \mathbb{E}[(\bar{Y})^2] \right)$$

$$= \boxed{\mathbb{E}[Y_1^2]} - \boxed{\mathbb{E}[(\bar{Y})^2]}$$

$$\mathbb{E}[Y_1^2] = \underbrace{\mathrm{Var}[Y_1] + (\mathbb{E}[Y_1])^2}_{= \sigma^2 + \mu^2}$$

$$\mathbb{E}[(\bar{Y})^2] = \underbrace{\mathrm{Var}[\bar{Y}] + (\mathbb{E}[\bar{Y}])^2}_{= \frac{\sigma^2}{n} + \mu^2}$$

$$\mathbb{E}[(S')^2] = \sigma^2 + \cancel{\mu^2} - \left(\frac{\sigma^2}{n} + \cancel{\mu^2} \right) = \left(1 - \frac{1}{n}\right) \sigma^2 = \boxed{\left(\frac{n-1}{n}\right) \sigma^2}$$

$$\Rightarrow \quad \mathrm{bias}((S')^2) = \mathbb{E}[(S')^2 - \sigma^2] = -\frac{\sigma^2}{n}$$

$$\mathbb{E}\left[(S')^2 \cdot \frac{n}{n-1}\right] = \sigma^2$$

$$\| $$

$$\mathbb{E}\left[\frac{\cancel{n}}{n-1} \cdot \frac{1}{\cancel{n}} \sum (Y_i - \bar{Y})^2\right]$$

$$= \sigma^2$$

So, the UNBIASED estimator for σ^2 is

$$\boxed{S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2}$$